

TOKKO, Russian Federation

WMO: 311370

Lat: 56.283N

Lon: 131.133E

Elev: 2788

StdP: 13.27

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-59.5	-56.0	-61.8	0.1	-58.8	-58.4	0.2	-55.9	11.5	-2.3	7.7	-18.0	0.0	50	0.758

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.1	80.9	62.9	77.2	61.6	73.4	59.7	65.2	76.4	63.2	73.6	61.3	70.9	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.4	90.0	69.3	59.3	83.5	67.6	57.3	77.6	65.9	31.8	76.6	30.2	73.5	28.7	71.2	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.1	7.9	DB	-63.6	85.4	5.1	3.7	-67.3	88.0	-70.3	90.2	-73.1	92.3	-76.8	95.0	(4)
(5)			WB	-63.2	69.4	5.2	3.5	-66.9	71.8	-70.0	73.9	-72.9	75.8	-76.7	78.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	-33.0	-24.3	-6.2	18.6	39.6	53.0	58.1	52.6	39.0	16.6	-14.9	-32.4	(6)
(7)		DBStd	34.82	11.24	10.41	11.81	11.11	7.46	6.08	5.53	6.10	7.94	11.42	14.84	11.03	(7)
(8)		HDD50	13629	2573	2080	1744	943	329	37	6	43	337	1035	1946	2556	(8)
(9)		HDD65	18589	3038	2500	2209	1393	786	362	220	385	779	1500	2396	3021	(9)
(10)		CDD50	523	0	0	0	0	7	126	258	124	8	0	0	0	(10)
(11)		CDD65	9	0	0	0	0	0	1	7	1	0	0	0	0	(11)
(12)		CDH74	633	0	0	0	0	5	129	392	105	2	0	0	0	(12)
(13)		CDH80	110	0	0	0	0	0	16	83	11	0	0	0	0	(13)
(14)	Wind	WSAvg	1.9	0.6	1.0	2.0	2.8	3.4	2.7	2.2	2.2	2.4	1.8	0.8	0.5	(14)
(15)	Precipitation	PrecAvg	18.6	0.4	0.3	0.4	1.1	1.9	2.7	3.4	3.5	2.3	1.4	0.7	0.5	(15)
(16)		PrecMax	25.9	0.9	0.7	1.1	2.6	4.5	4.5	8.4	7.1	5.0	3.4	1.3	1.1	(16)
(17)		PrecMin	12.2	0.1	0.1	0.0	0.1	0.3	0.7	1.1	0.9	0.4	0.4	0.1	0.2	(17)
(18)		PrecStd	3.0	0.2	0.2	0.2	0.6	0.9	0.9	1.6	1.6	1.1	0.7	0.3	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	14.0	34.7	53.9	72.8	82.6	87.8	81.8	69.9	48.5	28.8	5.9	(19)
MCWB			5.8	11.8	29.4	42.9	53.7	61.3	66.2	63.9	55.2	40.7	25.3	4.8	(20)	
2%		DB	-2.5	7.0	27.3	46.7	66.6	78.5	82.4	77.3	63.7	41.8	21.4	-2.5	(21)	
		MCWB	-3.1	5.4	23.4	38.1	50.9	60.6	64.1	62.5	52.3	36.4	19.8	-3.1	(22)	
5%		DB	-8.2	0.4	22.5	41.6	61.4	74.5	79.0	73.4	59.2	36.9	15.3	-9.0	(23)	
		MCWB	-8.6	-0.7	19.3	34.2	48.2	58.2	62.8	60.7	50.2	32.9	14.2	-9.5	(24)	
10%		DB	-13.9	-5.2	17.3	37.6	56.7	70.7	74.9	69.1	54.9	33.3	9.9	-14.0	(25)	
		MCWB	-14.3	-6.0	14.6	31.4	45.6	56.2	61.5	58.8	47.7	30.5	8.9	-14.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	11.9	29.8	43.8	55.0	65.0	69.1	66.9	58.1	41.4	26.0	4.8	(27)
MCDWB			7.7	13.8	33.9	52.6	69.0	76.5	81.2	77.9	66.0	47.6	28.6	5.5	(28)	
2%		WB	-3.0	5.4	23.6	38.3	51.8	62.1	66.3	64.2	54.1	36.9	19.7	-3.1	(29)	
		MCDWB	-2.5	6.9	27.3	45.9	64.6	74.8	78.4	74.4	61.1	41.3	21.5	-2.6	(30)	
5%		WB	-8.6	-0.7	19.3	34.7	49.0	59.9	64.2	61.9	50.8	33.5	14.2	-9.4	(31)	
		MCDWB	-8.1	0.2	22.6	40.7	59.6	71.1	75.0	70.7	57.6	36.4	15.2	-9.0	(32)	
10%		WB	-14.2	-6.0	14.7	31.8	46.3	57.7	62.5	59.6	48.2	30.6	8.9	-14.3	(33)	
		MCDWB	-13.9	-5.2	17.3	37.1	55.8	68.1	72.3	67.7	54.0	33.1	9.9	-13.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	25.3	33.3	36.5	29.2	23.7	27.8	27.1	26.8	23.0	22.7	26.8	22.0	(35)
MCDBR			26.9	35.4	37.7	31.2	35.6	38.3	38.2	37.1	32.8	24.7	26.5	22.6	(36)	
MCWBR			26.2	33.9	33.4	22.9	21.7	22.1	21.7	22.9	21.7	19.2	24.5	22.0	(37)	
5% WB		MCDBR	26.9	35.3	36.3	27.6	32.2	32.4	31.0	31.7	29.3	23.1	26.1	22.6	(38)	
		MCWBR	26.2	33.9	32.4	20.7	20.6	20.5	19.4	20.8	20.3	18.3	24.3	21.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.227	0.257	0.300	0.386	0.430	0.420	0.436	0.376	0.325	0.293	0.239	0.216	(40)
taud		2.101	2.110	2.034	1.869	2.022	2.213	2.194	2.368	2.430	2.254	2.088	2.035	(41)		
Ebn at Noon		217	256	270	258	255	261	253	264	265	239	205	185	(42)		
Edh at Noon		18	28	39	54	50	42	42	33	27	24	18	15	(43)		
(44)	All-Sky Solar Radiation	RadAvg		290	644	1177	1639	1609	1729	1586	1268	867	555	350	201	(44)
(45)		RadStd		16	33	54	98	146	128	139	121	88	59	19	8	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page